# 1. Problem Statement (Debate Agent)

In many academic, professional, and real-world scenarios, decision-making requires analyzing a topic from multiple perspectives. Debates are a powerful method to explore opposing viewpoints, evaluate arguments, and develop critical thinking skills. However, conducting a structured debate manually is time-consuming, requires multiple participants, and often lacks consistency in reasoning quality.

Students, researchers, and learners frequently need to understand both sides of an issue—such as ethical dilemmas, technological impacts, or policy decisions—but gathering balanced arguments, organizing them logically, and evaluating their strength requires significant effort. Additionally, traditional AI systems or single LLM responses tend to generate one-sided or blended answers, which do not accurately reflect the dynamic and adversarial nature of real debates.

The challenge, therefore, is to design an intelligent system that can simulate a structured debate by generating clear, logically consistent arguments from opposing perspectives, organizing them into a coherent format, and evaluating their effectiveness.

A single-step LLM approach is insufficient for this task because a debate inherently involves:

- Multiple independent viewpoints
- Sequential reasoning and interaction
- Role-based argument generation
- Evaluation and comparison of arguments

To address this, an **agentic architecture** is required, where multiple specialized agents are assigned distinct roles such as:

- a **Pro Agent** to support the topic
- a **Con Agent** to oppose the topic
- a **Moderator Agent** to structure the debate
- an **Evaluator Agent** to assess argument quality

By decomposing the problem into these interacting agents, the system can simulate realistic debates, improve reasoning quality, and provide structured, multi-perspective insights. This approach demonstrates how multi-agent systems can enhance decision-making and knowledge exploration compared to traditional single-model solutions.